USER INTERFACE AND DESIGN

EXPERIMENT – 10

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

index.html > html > head > style > body

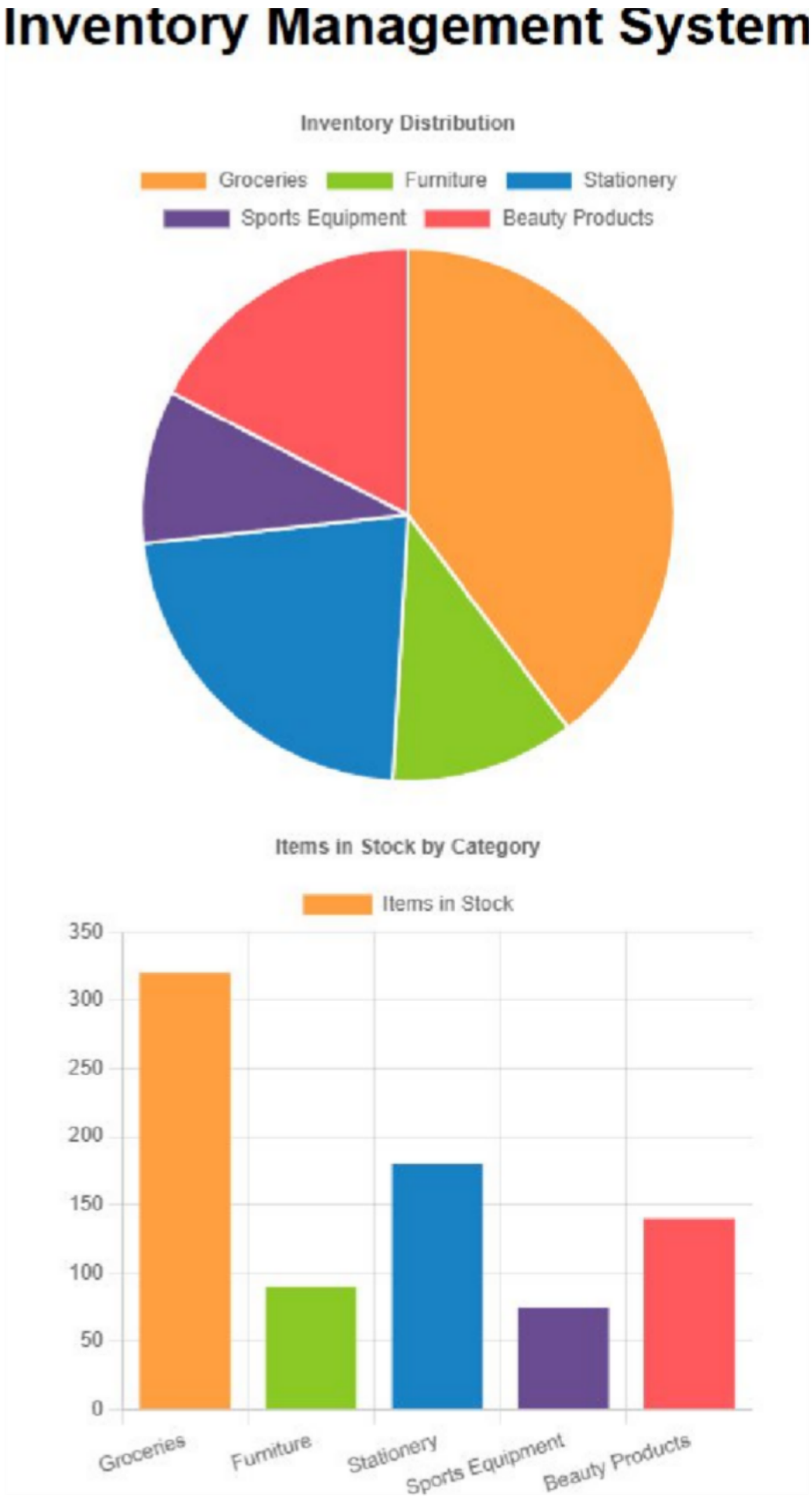
```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Inventory Management Visualization</title>
<style>
body {
font-family: Arial, sans-serif;
text-align: center;
margin: 50px;
}
canvas {
margin: 20px auto;
}

</style>
</head>
<body>
<h1>Inventory Management System</h1>
<canvas id="pieChart" width="400" height="400"></canvas>
<canvas id="barChart" width="400" height="400"></canvas>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>
</body>
</html>
```

Step 2:

Create the JavaScript File for Charts Next, create a JavaScript file (script.js) to handle the data visualization logic.

```javascript
JS script.js > ...
const inventoryData = {
  labels: ['Groceries', 'Furniture', 'Stationery', 'Sports Equipment', 'Beauty Products'],
  datasets: [
    {
      label: 'Items in Stock',
      data: [320, 90, 180, 75, 140],
      backgroundColor: [
        '#FF9F40',
        '#8AC926',
        '#1982C4',
        '#6A4C93',
        '#FF595E'
      ],
    }
  ]
};
// Creating the Pie Chart
const ctxPie = document.getElementById('pieChart').getContext('2d');
const pieChart = new Chart(ctxPie, {
  type: 'pie',
  data: inventoryData,
  options: {
    responsive: true,
    plugins: {
      title: {
        display: true,
        text: 'Inventory Distribution'
      }
    }
  }
});
// Creating the Bar Chart
const ctxBar = document.getElementById('barChart').getContext('2d');
const barChart = new Chart(ctxBar, {
  type: 'bar',
  data: inventoryData,
  options: {
    responsive: true,
    plugins: {
      title: {
        display: true,
        text: 'Items in Stock by Category'
      }
    },
    scales: {
      y: {
        beginAtZero: true
      }
    }
  }
});
```

OUTPUT:


Inventory Management System
Inventory Distribution
Groceries
Furniture
Stationery
Sports Equipment
Beauty Products
Items in Stock by Category
Items in Stock
350
300
250
200
150
100
50
0
Groceries
Furniture
Stationery
Sports Equipment
Beauty Products